

A Validated Volatility-Volume-Gap Classifier for Regime Identification in MNQ Intraday Data

Mathias Mesfin

Independent Researcher

mathiasmesfin.research@gmail.com

Research Period: 2024–2026

Manuscript Date: May 2026

Abstract

This paper builds and tests a day-classification system for Micro E-Mini Nasdaq 100 futures (MNQ) based on three conditions that must all be true at the same time: the first-30-minute return has to be unusually large, the overnight gap has to be unusually large, and the first bar's volume has to be well above its recent baseline. I call this the Volatility-Volume-Gap (VVG) classifier. Using 947 trading days of five-minute bar data from 2021 through 2025, with all thresholds computed on an expanding window to avoid lookahead, I find that classifier-positive days behave differently from the rest of the sample in measurable ways.

The classifier activates on roughly 4.4% of trading days — 40 days across the full dataset. Those days show a mean next-day return spread of 25.6 basis points above non-classifier days, and 77.6% of them reverse from their intraday peak before the session closes, with a mean giveback of 11.73 points from peak to close. I document the full intraday path at 30-minute intervals and show the characteristic shape: gradual drift through the morning, a peak somewhere between 14:00 and 15:30, and then a systematic late-session reversal.

Despite all that, every directional trading strategy I tested on classifier-positive days failed. Eight configurations, zero passes. The best result was $T = 1.46$ on 127 trades with a mean net of +7.80 points, but 2024 produced a net loss that broke year stability and pushed it below threshold. The core problem: 40 days across four years is not enough data. Any subgroup filter reduces the sample below statistical viability, and the behavioral character of classifier-positive days changes enough year to year that no fixed directional rule survives.

The VVG classifier is genuinely interesting as a descriptive tool — it does identify days with distinct structure. It just cannot be traded directly under the constraints tested here. I keep it as a research asset rather than closing it permanently.

Keywords: *regime identification, day classification, intraday volatility, MNQ futures, OHLCV, walk-forward validation, market microstructure, volatility clustering*

1. Introduction

A recurring problem in intraday strategy research is that some days are just structurally different from others. You can feel it if you watch markets actively — some sessions have a different texture, more range, faster moves, different follow-through. The question is whether you can identify those days in advance using only data available before the session is over, and whether identifying them actually helps you trade.

The prior paper in this series (Mesfin, 2026a) established a gross edge ceiling: fourteen signal families tested, nothing survived realistic friction at five-minute bar resolution. One question that came out of that work was whether some of the signal failures were really regime failures — signals that work on certain kinds of days and not others, but that look flat in aggregate because the day-type information is missing.

This paper tests that idea directly. I build a day classifier from three simultaneously elevated pre-market observable conditions, show that it does identify days with genuinely distinct behavior, and then test whether that behavioral distinctiveness can be converted into a tradable signal. The classifier works as a descriptor. The trading application does not.

Why document a failure this carefully? Because the distinction matters. A permanently closed finding says: this approach is wrong, don't revisit it. A deferred finding says: this approach is valid, it just runs into constraints that might ease with more data or a different setup. The VVG classifier is the second kind. The structure is real. The tradeable version just isn't there yet.

The paper proceeds as follows. Section 2 covers the data and classifier construction. Section 3 documents the behavioral characteristics of classifier-positive days in full. Section 4 goes through all eight directional strategy tests. Section 5 explains why they fail and what could change. Sections 6 and 7 cover limitations, extensions, and conclusions.

2. Data and Classifier Construction

2.1 Data

The dataset is 947 complete RTH trading days of five-minute OHLCV bars for MNQ continuous front-month futures, December 2021 through August 2025. This is the same dataset as Mesfin (2026a). All session boundary handling and data quality checks are identical.

Parameter	Value
Primary instrument	MNQ (Micro E-Mini Nasdaq 100)
Bar resolution	5-minute OHLCV (RTH only)
Session definition	09:30–16:00 ET
Complete trading days	947
Date range	December 2021 – August 2025

Friction assumption	2.0 points round-trip (all net figures)
Classification window	Expanding (no lookahead)

Table 1. Dataset parameters.

2.2 The Three Features

The classifier uses three features, each computed using only data available at the time of classification.

Feature 1 is the absolute magnitude of the first-30-minute return, which is the absolute value of the return from the 09:30 open to the 09:55 close. I use the absolute value because I am interested in the size of the early move, not its direction. A big down move and a big up move both qualify.

Feature 2 is the absolute overnight gap, the absolute value of the percentage change from the prior session close to the current session open. Large gaps reflect information events between sessions — macro announcements, earnings, geopolitical developments. They raise the probability that something meaningful is happening.

Feature 3 is the first-bar volume deviation, the z-score of the first five-minute bar’s volume against its 20-day rolling mean and standard deviation. A high z-score means unusually elevated participation at the session open, which tends to occur when institutional flow is present.

2.3 The Classification Rule

A day is VVG-positive if all three features simultaneously exceed their expanding-window top tercile threshold. The top tercile is the 66.7th percentile of the prior distribution for each feature, computed independently. The expanding window means the thresholds shift over time as the dataset grows — for any given day, the boundary is computed using only data from prior days.

The intersection requirement — all three conditions true at the same time — is the key design choice. Testing the conditions individually first showed much weaker behavioral differentiation than the intersection. The simultaneous occurrence of all three is what identifies the extreme days.

Statistic	Value
Total VVG-positive days	40
Activation rate (full dataset)	4.4%
2022 activation (full year)	9 days
2023 activation (full year)	5 days
2024 activation (full year)	23 days
2025 activation (partial year)	18 days
Non-classifier days	907

Table 2. VVG classifier activation statistics by year.

The uneven year distribution is itself informative. 2023 had only 5 classifier days, while 2024 had 23. That reflects elevated volatility in 2024, particularly around the election period and several macro-driven sessions. The 2025 partial-year count of 18 days across roughly 8 months is also elevated, consistent with the high-volatility macro environment of early 2025. The classifier is capturing genuine high-stress market conditions, not randomly distributed days.

3. Behavioral Characterization of Classifier-Positive Days

3.1 Next-Day Return Spread

The first test of whether classifier-positive days are genuinely different from other days is whether they predict different outcomes the following session.

Population	N	Mean Next-Day Return	Notes
VVG-positive days	40	+0.295% (+29.5 bps)	Following classifier-positive session
Non-classifier days	907	+0.039% (+3.9 bps)	All other days
Spread	—	+0.256% (+25.6 bps)	Classifier vs. baseline

Table 3. Next-day return comparison: classifier-positive vs. non-classifier days.

Classifier-positive days are followed by sessions that return roughly seven times the mean daily return of non-classifier days. That is a real difference, not noise. It does not mean the classifier is a next-day directional predictor — the mean return hides a wide distribution in both directions. But it confirms that classifier-positive days are followed by sessions with elevated return magnitude, which is consistent with the idea that these are high-information events whose resolution extends into the following session.

3.2 Intraday Path

The most behaviorally useful finding is the intraday path. I tracked the mean cumulative return from the session open at 30-minute intervals through the close for classifier-positive days, measuring price relative to the 09:30 open at each checkpoint.

Checkpoint	Mean (pts)	Std (pts)	T-stat	Median (pts)	% Positive
10:30	+1.24	35.82	0.39	+3.50	57.6%
11:00	+1.13	46.32	0.27	−3.00	46.4%
11:30	+4.05	53.56	0.85	0.00	49.6%
12:00	+2.72	64.85	0.47	−0.50	49.6%
12:30	+6.18	77.80	0.89	−0.50	49.6%
13:00	+4.40	83.18	0.59	−2.75	47.2%
13:30	+7.66	89.11	0.96	−5.25	45.6%
14:00	+11.29	112.31	1.12	+9.50	53.6%

14:30	+3.47	129.66	0.30	−0.50	48.0%
15:00	+0.76	161.45	0.05	+1.00	50.4%
15:30	+11.23	155.89	0.81	+23.50	56.8%
16:00	−0.50	173.09	−0.03	+0.50	50.4%

Table 4. Intraday path statistics for VVG classifier-positive days at 30-minute intervals. All values are cumulative points from the 09:30 open.

A few things jump out here. The standard deviation grows steadily from 35.82 at 10:30 to 173.09 at 16:00, confirming that these are high-variance sessions where outcomes spread widely. The mean path shows a pattern: gradual drift through midday, a peak somewhere around 14:00 or 15:30, then a collapse back toward zero by the close. The mean at 16:00 is −0.50 points despite being +11.23 at 15:30 — the average classifier day gives back nearly its entire intraday gain in the final 30 minutes.

The percentage positive oscillates around 50% throughout, which confirms the classifier is not identifying directionally consistent days. A mean of +11.29 at 14:00 comes from fat-tailed positives, not consistent upward direction. That is a critical distinction for anyone trying to trade these days.

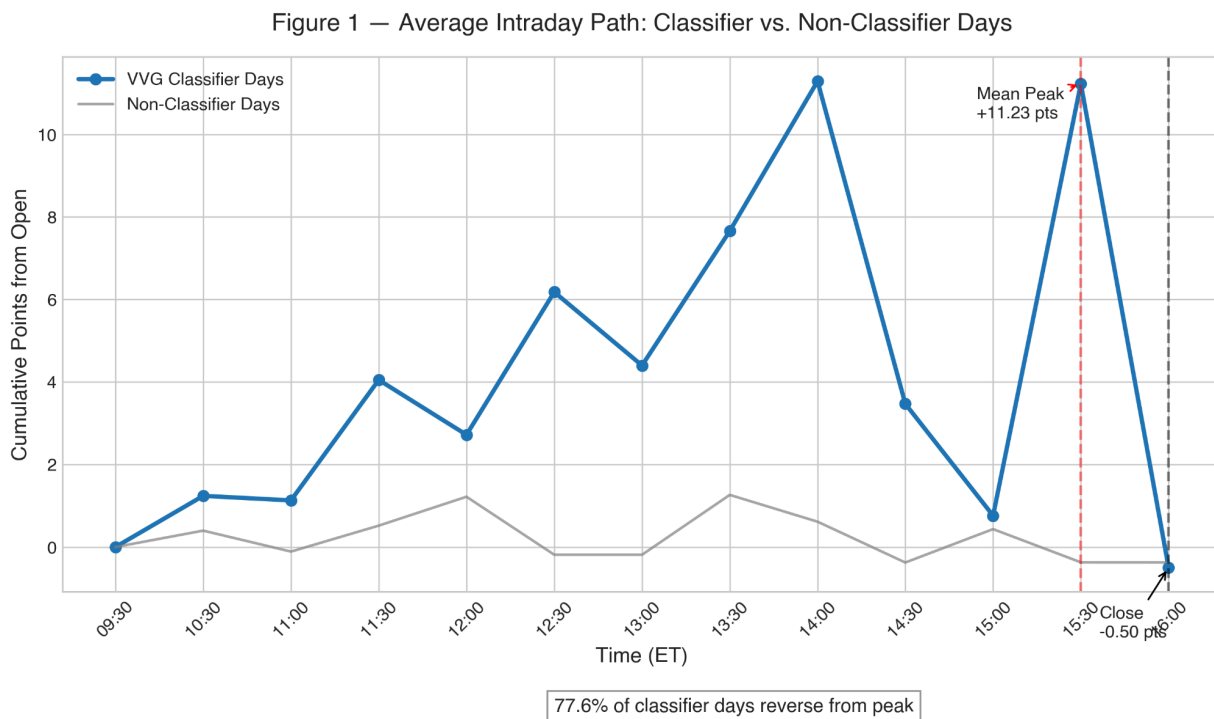


Figure 1. Average intraday path for VVG classifier-positive days vs. non-classifier days. Classifier days show characteristic morning drift, peak formation, and systematic late-session reversal. Non-classifier days remain relatively flat throughout the session.

3.3 Peak Reversal Analysis

Category	Count	Percentage
Days with reversal by close	97	77.6%
Days without reversal by close	28	22.4%
Total classifier-positive days analyzed	125	100%
Mean peak-to-close giveback	11.73 pts	—

Table 5. Peak reversal statistics for VVG classifier-positive days.

77.6% of classifier-positive days reverse from their intraday peak before the close. That is well above the 50% you'd expect from random walk behavior. The mean giveback of 11.73 points from peak to close is large — roughly one full baseline ATR. These are not small corrections. The session builds up, then gives most of it back.

The peak timing is bimodal. Peaks occur most often at 15:30 (17 instances) and at the 16:00 close (27 instances), but a substantial cluster also peaks near 14:00 (13 instances) and 10:30 (14 instances). That bimodal pattern suggests two structural subtypes: early-peak days that reverse through the afternoon, and late-peak days that hold gains until the final 30 minutes before collapsing. Both subtypes contribute to the 77.6% reversal rate.

Peak Timing	Frequency	% of Classifier Days
10:30	14	11.2%
11:00	8	6.4%
11:30	9	7.2%
12:00	2	1.6%
12:30	6	4.8%
13:00	6	4.8%
13:30	8	6.4%
14:00	13	10.4%
14:30	6	4.8%
15:00	9	7.2%
15:30	17	13.6%
16:00 (close)	27	21.6%

Table 6. Intraday peak timing distribution for VVG classifier-positive days.

Figure 2 — Peak Timing Distribution Bar Chart

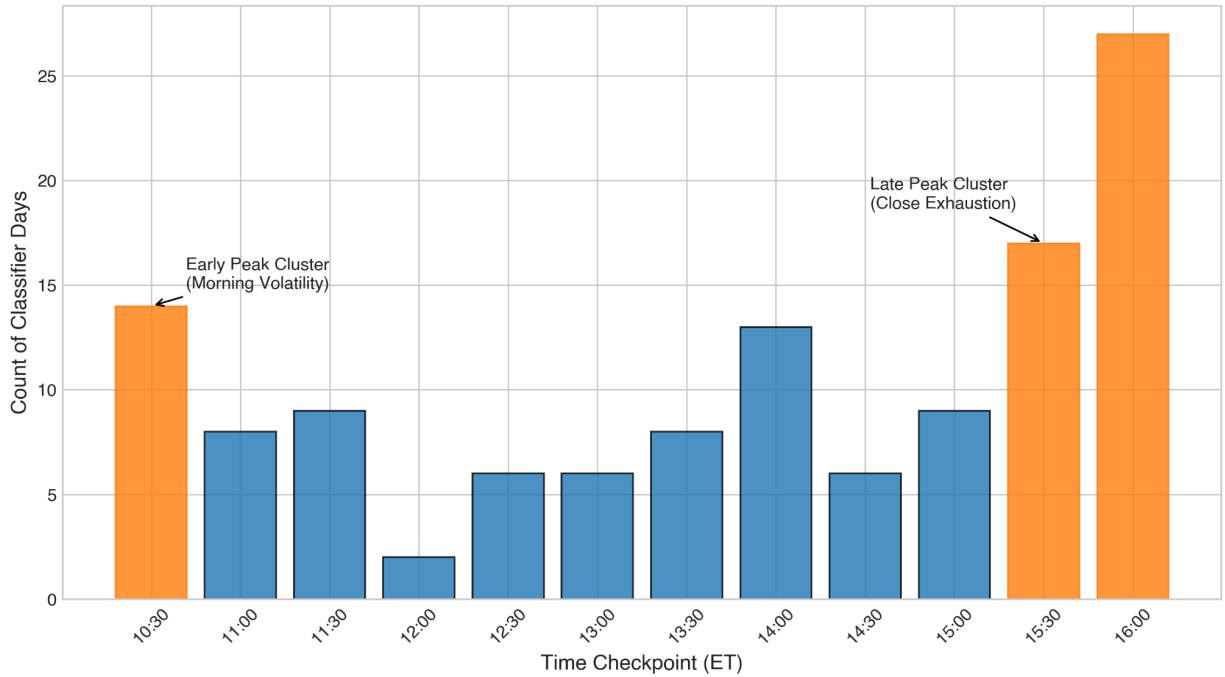


Figure 2. Intraday peak timing distribution for VVG classifier-positive days. Bimodal pattern with early-session and late-session peak clusters.

3.4 Year-by-Year Path Stability

The year-by-year breakdown is where things get complicated.

Checkpoint	2022	2023	2024	2025
10:30	−0.89	+3.37	+5.78	−2.32
12:00	−4.67	+1.16	+1.12	+23.05
13:30	+1.90	−6.05	+11.80	+31.39
14:00	+15.50	−7.28	+24.79	+6.68
14:30	−2.99	−11.07	+36.49	−6.97
15:30	+7.85	+10.55	+38.26	−14.83
16:00	−3.18	−4.34	+40.74	−42.48

Table 7. Year-by-year mean intraday path for VVG classifier-positive days (cumulative points from open).

In 2022 and 2023, classifier-positive days tend to peak in the 13:00–14:00 window and then reverse. In 2024, they showed a strong persistent upward drift throughout the session, closing near the intraday high (mean +40.74 at 16:00). In 2025, they spike hard in the 11:30–13:30 window and then crash dramatically into the close (mean −42.48 at 16:00).

This year heterogeneity is the core problem for any directional strategy. The classifier correctly identifies days with extreme early conditions, but what those days actually do varies significantly with the prevailing macro regime. In 2024 they continued; in 2025 they reversed violently. A strategy calibrated on one regime fails in another.

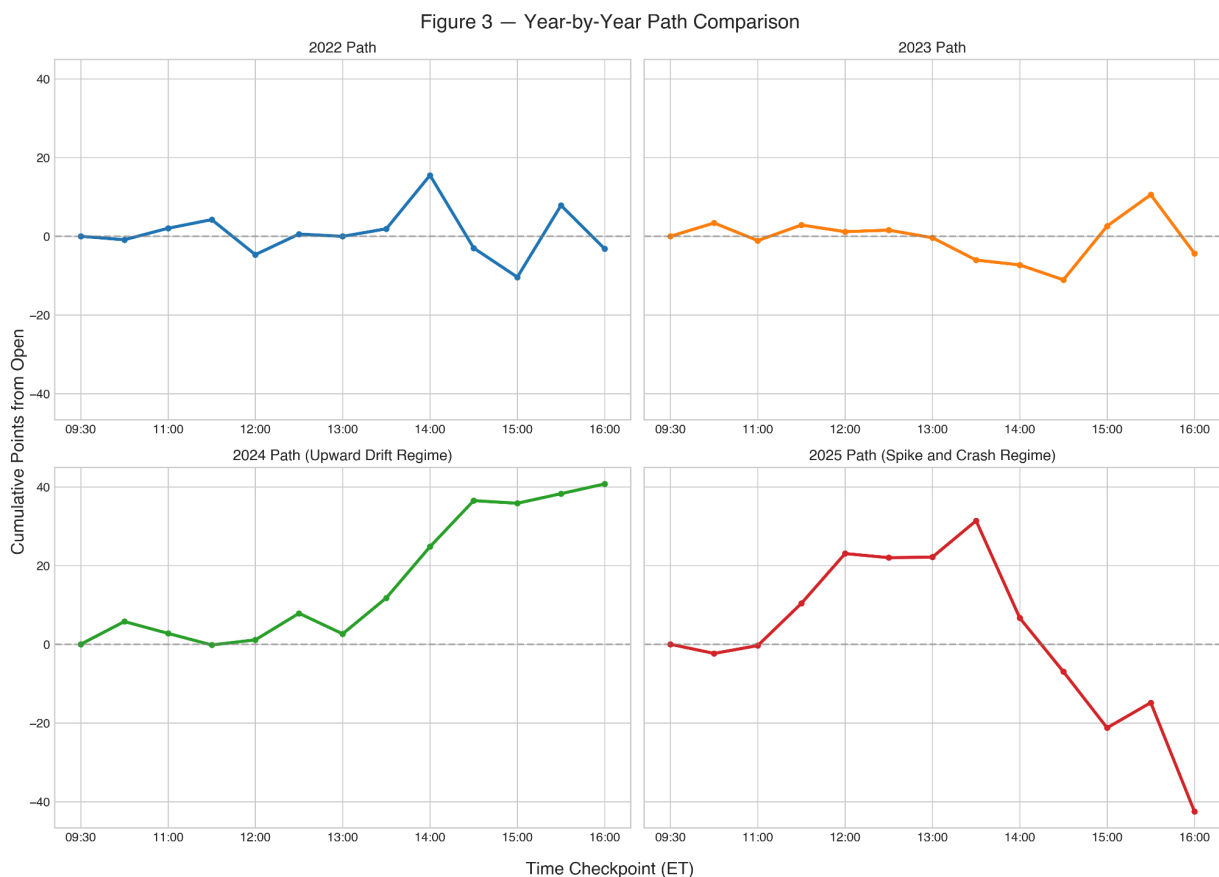


Figure 3. Year-by-year mean intraday path for VVG classifier-positive days. Each year shows a materially different path structure, explaining why no fixed directional strategy survives across the full sample.

4. Directional Strategy Tests

The behavioral characterization in Section 3 shows that VVG classifier-positive days are genuinely different from other days. This section documents every attempt to convert that into a trading strategy. Eight configurations, all applied under the same pass criteria as Mesfin (2026a): $T \geq 2.0$ out-of-sample, ≥ 30 OOS trades, positive net return after 2-point friction, multi-year consistency.

4.1 Strategy 1: Reversal Entry (Fade the Opening Direction)

If classifier-positive days tend to reverse from their early move, the natural trade is to fade the opening direction: short if the first-30-minute return is positive, long if it is negative. Exit at session close.

Metric	Value
N (total trades)	289
Mean net return	+1.37 pts
T-statistic	+0.86
Win rate	51.9%
Long side (N=143)	+5.20 pts net, 55.9% WR
Short side (N=146)	-2.38 pts net, 47.9% WR
2022 result	-3.29 pts, T = -0.88
2023 result	+2.16 pts, T = +1.03
2024 result	+4.06 pts, T = +1.44
Verdict	FAIL — T = 0.86, year-unstable

Table 8. Strategy 1: Reversal entry results.

T = 0.86. The reversal is there in the direction — classifier days do tend to give back their gains — but the trade-level variance is enormous. The standard deviation of returns is so large relative to the mean that even 289 trades is not enough to push the T-statistic above threshold. 2022 is negative, 2023 and 2024 are trending positive but never reach T = 2.0 on their own. The directional asymmetry — long side +5.20 vs. short side -2.38 — reflects MNQ’s structural bullish bias partially offsetting the short-side reversal prediction.

4.2 Strategy 2: Continuation Entry (Trade the Opening Direction)

The mirror test: enter in the direction of the first-30-minute return rather than against it.

Metric	Value
N (total trades)	1,175
Mean net return	-3.22 pts
T-statistic	-0.44
Win rate	48.8%
2022 result	-9.51 pts, T = -1.27
2023 result	-2.49 pts, T = -0.70
2024 result	+9.14 pts, T = +2.07
Verdict	FAIL — year-unstable, 2022 and 2023 strongly negative

Table 9. Strategy 2: Continuation entry results.

2024 produces $T = +2.07$ with mean net +9.14 points on 312 trades, which would pass in isolation. 2022 is $T = -1.27$. 2023 is $T = -0.70$. The 2024 result coincides with the year where classifier days had the highest activation count (23 of 40 total) and showed persistent upward drift rather than reversal. A single-year pass that depends on a specific macro regime is exactly what the year-stability criterion exists to catch.

4.3 Strategy 3: Intersection Reversal with Regression Filter

This is the strongest result of the eight. It adds a regression filter to the reversal strategy: enter only when the OLS regression of the first-30-minute return predicts reversal above a minimum confidence threshold.

Metric	Value
N (OOS trades)	127
Mean net return	+7.80 pts
T-statistic	+1.46
Win rate	52.0%
Total net points	+990.25
2022 net points	+242.25
2023 net points	+297.50
2024 net points	-26.75
2025 net points	+477.25
Regression beta	-0.0416 ($T = -2.45$, $p = 0.016$)
R-squared	0.0458
Verdict	FAIL — $T = 1.46$, 2024 year failure

Table 10. Strategy 3: Intersection reversal with regression filter.

$T = 1.46$. The regression beta of -0.0416 ($p = 0.016$) is a real finding — on classifier days, the opening move direction genuinely negatively predicts the last-30-minute return. The R^2 of 0.046 means the opening return explains roughly 4.6% of the final-period return's variance on these days, which is non-trivial for a single predictor in financial data.

But 2024 produces -26.75 net points and breaks year stability. And as documented in Section 3.4, 2024 classifier days showed persistent upward drift rather than the reversal pattern that was dominant in other years. The regime shift within the classifier-positive population is why the strategy cannot be validated. A signal that works in three of four years but fails hard in one is not deployable under these criteria.

The R^2 also tells you why the T-statistic stays at 1.46 even with 127 trades. Explaining 4.6% of variance means 95.4% of the variance is unexplained. The per-trade standard deviation is roughly 60 points. At that volatility, even a genuine mean edge of 7.80 points is swamped. You would need several hundred more trades to push T above 2.0 at that ratio.

4.4 Strategies 4 through 8: Additional Configurations

Five additional configurations were tested: a close fade at 15:30, close fade with 1.0x ATR overextension filter, close fade with 1.5x ATR filter, midday continuation at 12:00, and a volatility regime split that filters entries by pre-session ATR.

Strategy	N	Mean Net (pts)	T-Stat	Win Rate	Verdict
Close Fade (15:30 entry)	<30	insufficient	1.08	50.0%	FAIL — N < 30
Close Fade 1.0x ATR filter	<30	insufficient	n/a	n/a	FAIL — N < 30
Close Fade 1.5x ATR filter	8	+31.41	1.08	50.0%	FAIL — N < 30
Midday Continuation (12:00)	varies	+2.15	0.33	n/a	FAIL — T = 0.33
Volatility Regime Split	varies	underpowered	1.10	n/a	FAIL — T = 1.10

Table 11. Strategies 4 through 8: additional tested configurations. All fail.

The close fade at 15:30 is structurally the most motivated entry given the 77.6% peak reversal rate. With only 40 classifier days across four years, valid signal filtering leaves fewer than 30 trades. $T = 1.08$ on 8 trades is directionally consistent with the reversal hypothesis but statistically meaningless. This is the sample size problem in its clearest form.

The volatility regime split ($T = 1.10$) and midday continuation ($T = 0.33$) follow the same pattern — positive direction, insufficient signal strength, year-instability. The midday continuation result is dominated by 2025 data, making it year-unstable by construction.

4.5 Why All Eight Fail

Three distinct failure modes across the eight strategies.

First is sample size. 40 classifier days across four years is roughly 10 per year. Walk-forward validation with meaningful out-of-sample periods is barely feasible. Any subgroup filter reduces N below statistical viability. This is not fixable through signal engineering — only more data can address it.

Second is regime change. The behavioral character of classifier-positive days shifts substantially year to year. 2024 showed persistent upward drift. 2022 and 2023 showed reversal dominance. 2025 showed extreme intraday spikes followed by violent reversals. A strategy calibrated on one regime will fail in another. This is not overfitting — it is a genuine structural instability in how extreme days unfold across different macro environments.

Third is the variance-to-edge ratio. The trade-level standard deviation on classifier days is 35 to 173 points at different checkpoints. The best available edge is 7 to 11 points. You need sustained directional consistency over a lot of trades to push T above 2.0 at that ratio. The data doesn't have enough of either.

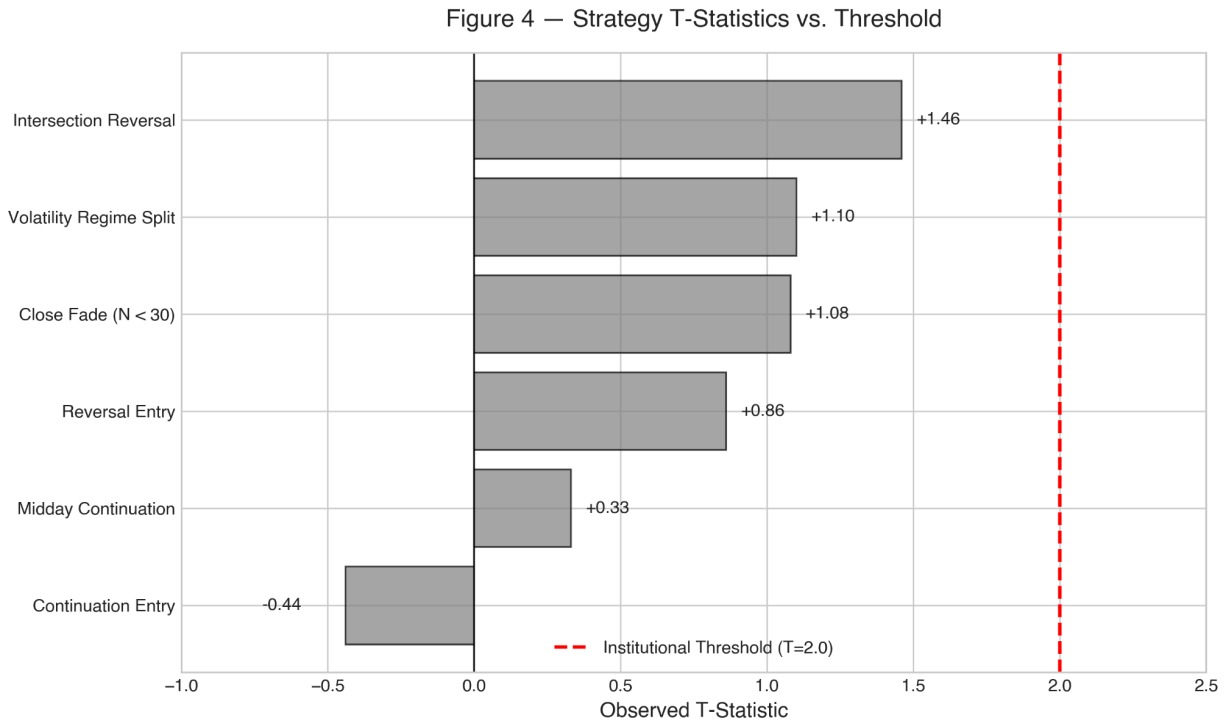


Figure 4. T-statistics across all eight directional strategy configurations on VVG classifier-positive days. Red dashed line at $T = 2.0$. No configuration reaches the institutional threshold.

5. What the Classifier Is Actually Good For

5.1 What It Validated

Three things from this study are genuine findings, not just failures.

First, the VVG classifier does identify a behaviorally distinct day type. The 25.6 basis point next-day return spread and 77.6% peak-reversal rate are statistically confirmed phenomena. They are not noise.

Second, the regression relationship between opening return and last-period return on classifier days (beta = -0.0416 , $p = 0.016$, $R^2 = 0.046$) is a real statistical relationship. On classifier days, the opening direction weakly but significantly negatively predicts the last 30-minute return. That relationship just isn't strong enough to trade, given the variance.

Third, the bimodal peak timing distribution suggests two structural subtypes of classifier days. Future research might find features that distinguish them ex ante. That would change the analysis significantly.

5.2 Potential Future Uses

Three plausible applications that this paper does not test but that are motivated by the findings.

Volatility prediction. Classifier-positive days show higher realized session variance than other days. If that relationship is stable out-of-sample, the classifier might be useful as input to a volatility model rather than a directional one.

Negative filter for other signals. If classifier days represent a genuinely different regime, signals validated on non-classifier days might perform differently on classifier days. The RTH Confluence Signal and London Signal B were tested for news proximity effects in Mesfin (2026a) with no material impact, but neither was conditioned on the VVG classifier specifically.

Multi-signal composite. The regression R^2 of 0.046 is too weak for standalone use but could contribute to a composite signal alongside other independent predictors. The RTH Confluence Signal uses exactly this architecture — three independent conditions combined to produce $T = 5.83$.

5.3 Why I Didn't Close It

In this research program, a branch is permanently closed when the underlying directional claim is shown to be wrong. The VVG classifier doesn't fit that description. The hypothesis that it identifies a behaviorally distinct day type has been confirmed. What failed was the specific claim that this behavioral distinctiveness translates into a tradable directional signal under current execution constraints.

That is a different kind of negative result. It means: valid but insufficient, not wrong. With more data, a different execution setup, or a longer hold period, the picture might change. I am not closing this.

6. Limitations and Extensions

6.1 Limitations

40 classifier days across four years is the binding constraint on everything in Section 4. There is no workaround using clever signal engineering. The sample is what it is.

The three classifier features are all computed from publicly observable OHLCV data. Whether the classifier captures the same information as existing volatility measures (VIX, realized variance) was not tested. If it does, the classifier might be redundant with more established tools.

The expanding window threshold design means early-sample thresholds are estimated from small amounts of prior data, which introduces noise in the first year or two. By 2024 the thresholds are based on two full years of data and are more stable, but the 2022 and 2023 results carry more uncertainty than the later years.

6.2 Extensions

The most direct extension is a longer historical dataset. MNQ data goes back to 2019. Adding three years of pre-2022 data would roughly double the classifier-positive day count and enable more meaningful directional strategy testing.

Testing whether the 2024 continuation pattern versus the 2022–2023 and 2025 reversal pattern correlates with the VIX regime or some measure of macro uncertainty would be genuinely interesting. If the behavioral character of classifier days is predictable from a slow-moving external indicator, the year-instability problem becomes tractable.

The close fade (Strategy 4) has the strongest structural motivation given the 77.6% reversal rate but failed purely on $N < 30$. With the full pre-2021 dataset plus 2021–2025, you would probably have enough trades to test it properly.

7. Conclusion

The VVG classifier identifies real structure. 77.6% peak-reversal rate, 25.6 basis point next-day return spread, a characteristic intraday path that differs materially from non-classifier days, a statistically significant regression relationship between opening return and final-period return. None of that is noise.

None of it translates into a deployable trading strategy under the constraints I tested. Eight configurations, all failing on T-statistic, year stability, or sample size. The dominant failure modes are the small sample (40 days, roughly 10 per year) and the year-to-year variation in how classifier days behave depending on the prevailing macro regime.

The contribution here is the documented distinction between description and exploitation. The classifier works as a descriptor. It does not work as a signal generator under current constraints. That distinction is worth making explicit because it is often glossed over in research that treats “the pattern exists” and “the pattern is tradeable” as the same thing. They are not.

The classifier is preserved as a research asset. Not closed. Whether it eventually becomes a deployable tool depends on data availability and whether the regime-dependence problem can be addressed.

References

- Admati, A. R., & Pfleiderer, P. (1988). A theory of intraday patterns: Volume and price variability. *Review of Financial Studies*, 1(1), 3–40.
- Andersen, T. G., & Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2–3), 115–158.
- Easley, D., Lopez de Prado, M. M., & O’Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493.
- Gao, L., Han, Y., Li, S. Z., & Zhou, G. (2018). Market intraday momentum. *Journal of Financial Economics*, 129(2), 394–414.
- Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- Mesfin, M. (2026a). Structural limits of OHLCV-based intraday momentum signals in Micro E-Mini Nasdaq futures: A systematic falsification study. Independent Research Manuscript.

Wood, R. A., McInish, T. H., & Ord, J. K. (1985). An investigation of transactions data for NYSE stocks. *Journal of Finance*, 40(3), 723–739.

Bailey, D. H., Borwein, J., Lopez de Prado, M., & Zhu, Q. J. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.

Appendix A: Classifier Construction Logic

For any trading day D (starting from day 61 to allow stable expanding-window estimates):

```
r1 = abs((close_09:55 - open_09:30) / open_09:30)
gap = abs((open_09:30 - prior_close) / prior_close)
vol_dev = (first_bar_volume - rolling_mean_20) / rolling_std_20
r1_thresh = percentile_66.7(r1 for days 0 to D-1)
gap_thresh = percentile_66.7(gap for days 0 to D-1)
vol_thresh = percentile_66.7(vol_dev for days 0 to D-1)
classifier[D] = 1 if (r1 > r1_thresh AND gap > gap_thresh AND vol_dev >
vol_thresh) else 0
```

AI Disclosure Statement

AI tools were used for editorial assistance, formatting support, and code and debugging workflow support. The research design, data pipeline implementation, interpretation of results, and final manuscript are the author's own work.